

Credit Risk Prediction Model

ID/X PARTNERS - DATA SCIENTIST

Presented by
Kuntur Jalassuad



Kuntur Jalassuad

Junior Data Scientist | Data Analyst Associate | Data and Information Processor

Data Analyst Associate with nearly 8 years of experience supporting data management, processing, and validation of large-scale institutional datasets. Skilled in using SQL, Python, and Google BigQuery to query data, support database optimization, and generate actionable insights for business needs. Experienced in maintaining data quality across thousands of academic records and developing clear data visualizations for management reporting. A graduate of a Data Science Bootcamp where I was awarded Most Outstanding Student and Best Final Project in Data Analysis. Hands-on experience includes building machine learning models for employee attrition prediction and developing interactive sales performance dashboards. Eager to continue growing as a Junior Data Scientist by applying strong data fundamentals and analytical thinking to solve real-world business problems.



East Jakarta, DKI Jakarta



jalassuad.k@gmail.com



<https://www.linkedin.com/in/kuntur-jalassuad/>

Courses and Certification

Data Science : Machine Learning | [<link certificate>](#)

August, 2025 – February, 2026

Data Analysis: Fullstack Intensive Bootcamp | [<link certificate>](#)

August– September, 2025

TOEFL ITP Test – Score 510 | [<link certificate>](#)

September, 2025

Microsoft Excel Basic to Advanced | [<link certificate>](#)

July – August, 2025

Mini Course Data Analytics | [<link certificate>](#)

June, 2025

Training Course: Legal Non For Legal | [<link certificate>](#)

December , 2023

About Company



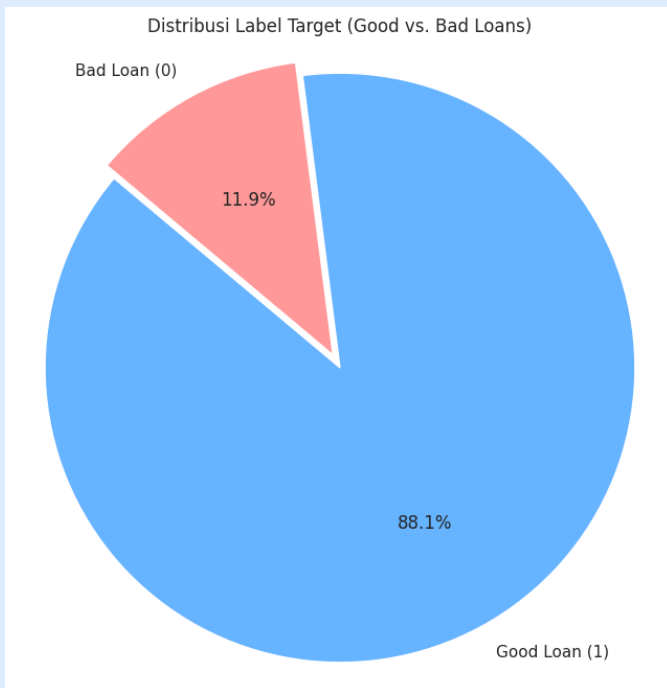
id/x partners is an Indonesian consulting firm, specializing in Data Analytics, Decisioning and RegTech solutions since 2006. We bring proven success stories in helping 85+ institutions - mostly top banks, multifinance companies, fintechs and insurers - leverage advanced analytics and AI-based solutions to solve challenges in digital lending, risk management, financial crime, and regulatory compliance.

- **Official Website:** idxpartners.com.
- **Professional Networking:** [id/x partners LinkedIn](#).

The company provides digital transformation and AI-driven solutions across several domains:

- **Financial Technology:** Digital lending, credit scoring, and predictive modeling.
- **Risk & Compliance:** Risk rating, risk management, and financial crime prevention.
- **Data Solutions:** Big data, data warehousing, and machine learning (AI/ML).
- **Regulatory Support:** Specialized solutions for IFRS9 and IFRS17 compliance.

BUSINESS PROBLEM — WHY THIS MATTERS



Problem

Our current Bad Loan rate (11.9%) is more than 2x higher than the OJK/Regulator's maximum threshold of 5%.

Impact

This "Bad Loan" level indicates a systemic failure in credit underwriting and poses a direct threat to our liquidity and capital adequacy

Risk

Continuous operation at this level risks regulatory sanctions and a total erosion of institutional profitability.



Link code [here](#)!



Project explanation video [here](#)!



Link Github : [here](#)

Data Understanding

— *Risk Landscape*



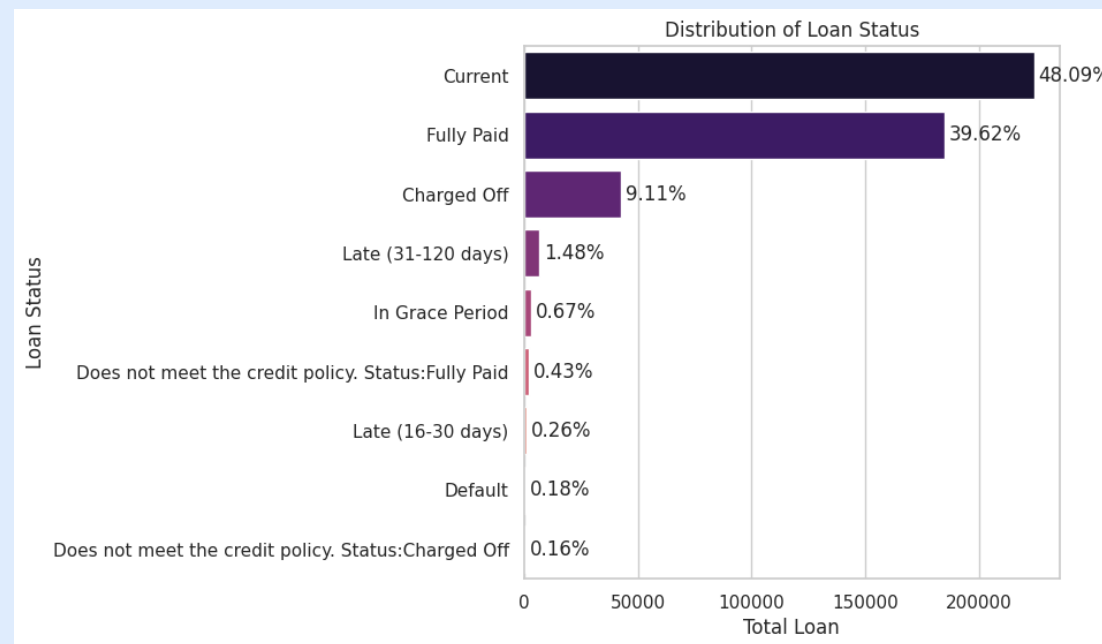
1. DATA UNDERSTANDING

Dataset : loan_data_2007_2014.csv

Data Shape	466285	Rows
	75	Feature

**Missing Values :
40 out of 75 columns**

Missing Value (%)	Sum of Feature	%
100%	17	23%
100% < x > 30%	5	7%
<30%	18	24%
0%	35	47%



Categories	Loan Status Distribution	Count Value
Bad	Charged Off	42,475
Bad	Late (31-120 days)	6,900
Bad	In Grace Period	3,146
Bad	Late (16-30 days)	1,218
Bad	Default	832
Bad	Does not meet the credit policy. Status:Charged Off	761
Total (Bad Status)		55,332 (11.86%)

WHAT IS OUR GOAL ?

Objective

Develop a machine learning model to predict credit risk to optimize business decisions and reduce potential losses.

Methodology

Executed end-to-end stages including Data Understanding, Exploratory Data Analysis (EDA), Data Preparation, Data Modelling, and Evaluation.

Modeling Approach

Implemented a minimum of five machine learning algorithms, with Logistic Regression as the mandatory baseline model.

Evaluation Metrics

Assessed model performance utilizing accuracy, precision, recall, ROC-AUC, F1 Score, and F2 Score.

EDA —

What Drives Credit Risk ?



2. EXPLORATORY DATA ANALYSIS (EDA)

STATISTIC CATEGORIAL

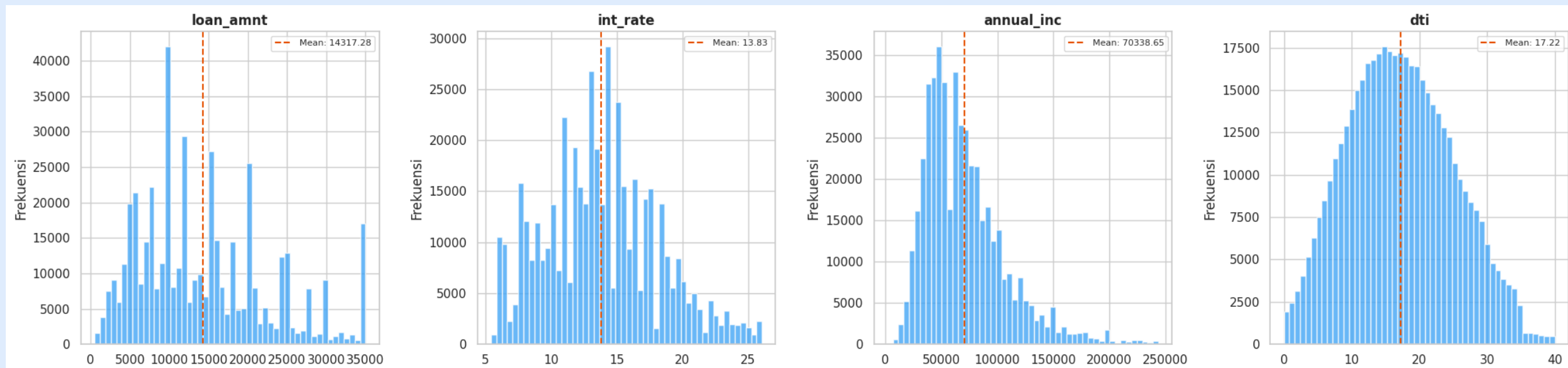
	count	unique	top	freq
term	466285	2	36 months	337953
grade	466285	7	B	136929
sub_grade	466285	35	B3	31686
emp_title	438697	205475	Teacher	5399
emp_length	445277	11	10+ years	150049
home_ownership	466285	6	MORTGAGE	235875
verification_status	466285	3	Verified	168055
issue_d	466285	91	Oct-14	38782
loan_status	466285	9	Current	224226
pymnt_plan	466285	2	n	466276
url	466285	466285	https://www.lendingclub.com/browse/loanDetail...	1
desc	125981	124435		234
purpose	466285	14	debt Consolidation	274195
title	466264	63098	Debt consolidation	164075
zip_code	466285	888	945xx	5304
addr_state	466285	50	CA	71450
earliest_cr_line	466256	664	Oct-00	3674
initial_list_status	466285	2	f	303005
last_pymnt_d	465909	98	Jan-16	179620
next_pymnt_d	239071	100	Feb-16	208393
last_credit_pull_d	466243	103	Jan-16	327699
application_type	466285	1	INDIVIDUAL	466285

INSIGHT :

1. Most loans are issued with **36-month terms (~72%)**, indicating borrower preference for shorter-to-medium loan duration with lower long-term risk.
2. The portfolio is dominated by **Grade B**, especially **Sub-grade B3**, reflecting a mid-risk borrower segment.
3. Most borrowers report **10+ years** of employment, suggesting strong income stability and lower default probability.
4. **Mortgage** is the most common home ownership type, indicating borrowers have assets along with long-term debt commitments.
5. The **Verified** status dominates, improving income data reliability for credit risk and underwriting models.
6. **Debt Consolidation** is the primary loan purpose, indicating borrowers aim to restructure or reduce existing debt burden.
7. **Current** is the most frequent loan status, suggesting a relatively healthy active loan portfolio.
8. The largest borrower concentration comes from **California (CA)**, with loan issuance clustered around **2014–2016**.
9. **emp_title** has very high cardinality → requires cleaning or NLP preprocessing, **desc** contains many missing values → potential source of bias

2. EXPLORATORY DATA ANALYSIS (EDA)

Univariate Analysis



The loan amount distribution is **right-skewed**, where most loans fall in mid-range values and only a small portion are very large loans.

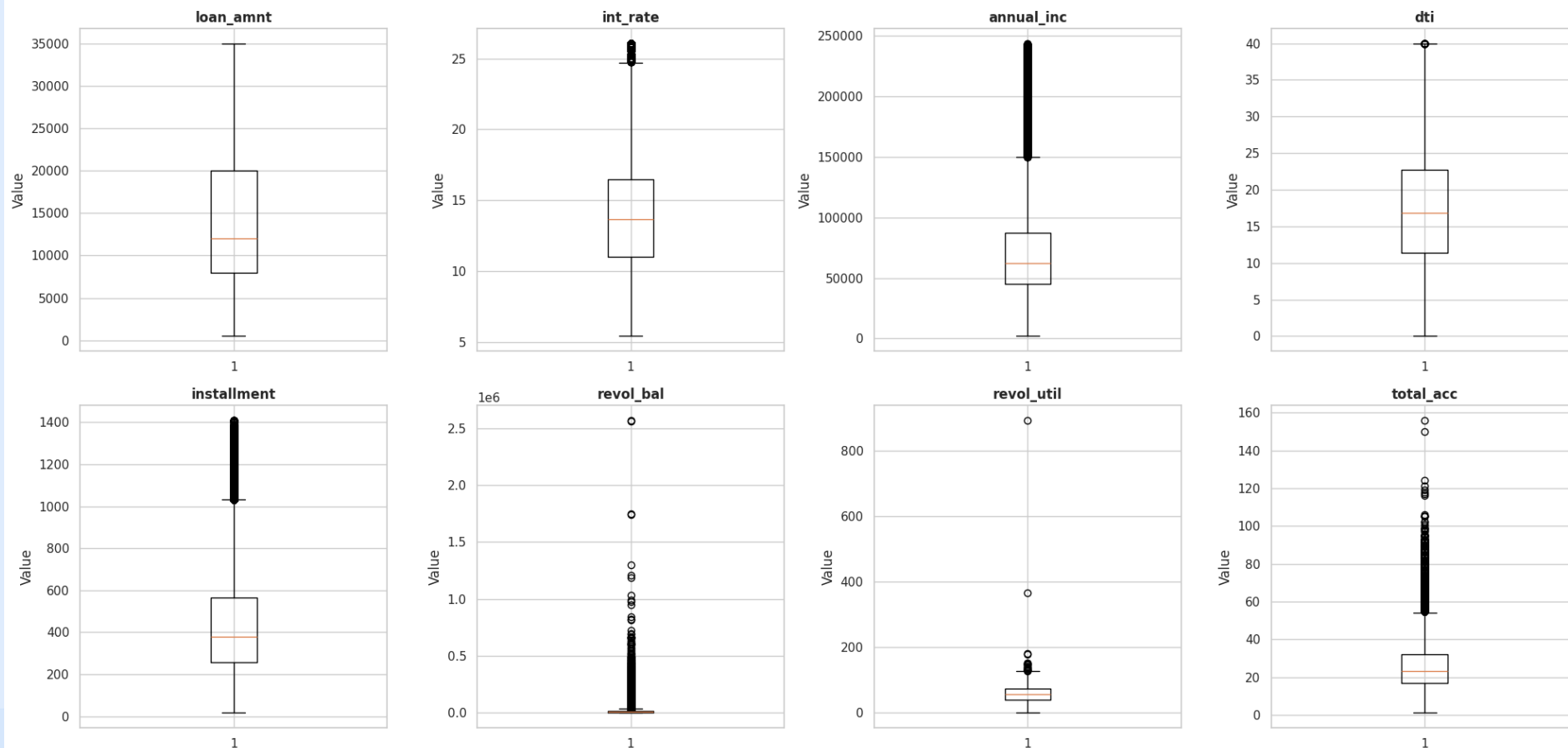
Interest rates are concentrated around **10–15%**, forming a near-normal distribution with slight right skew.

Annual income shows a **strong right-skew**, with most borrowers in middle-income ranges and a few extreme high-income outliers.

DTI displays a **near-normal distribution**, mostly between **10–20%**.

2. EXPLORATORY DATA ANALYSIS (EDA)

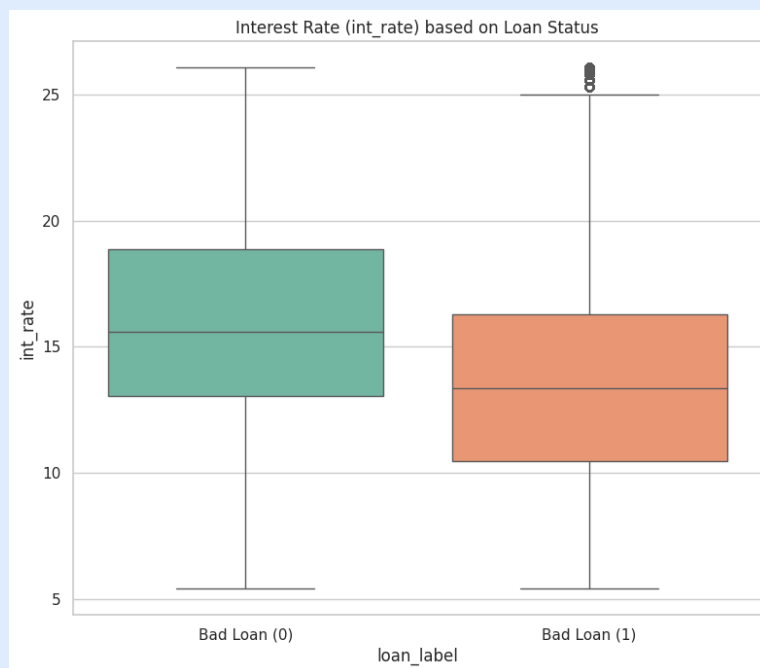
Boxplots of Numerical Variables



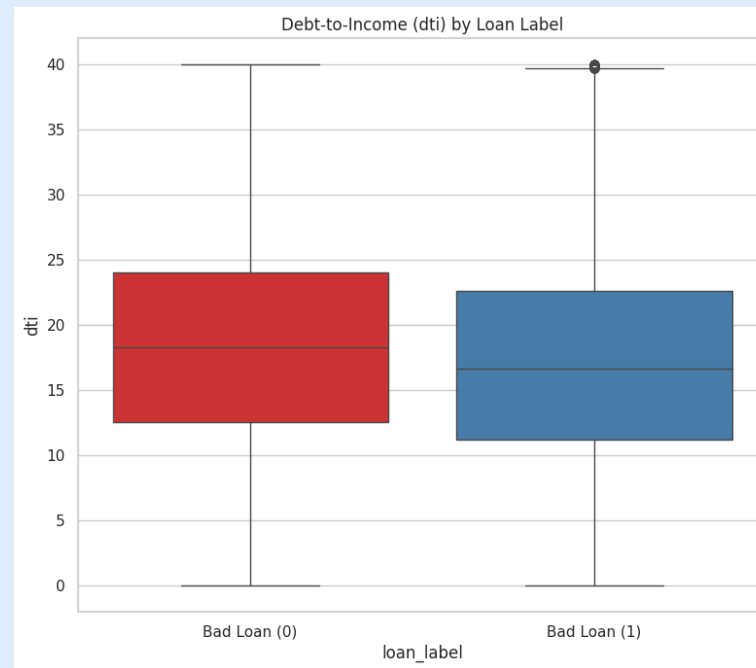
- Most numerical variables exhibit **outliers and right-skewness**, especially financial balance and income features.
- **Outlier treatment and data transformation** are essential steps **before modeling**.
- Variables such as **int_rate**, **dti**, and **revol_util** remain **highly informative for credit risk analysis**.

Univariate Analysis

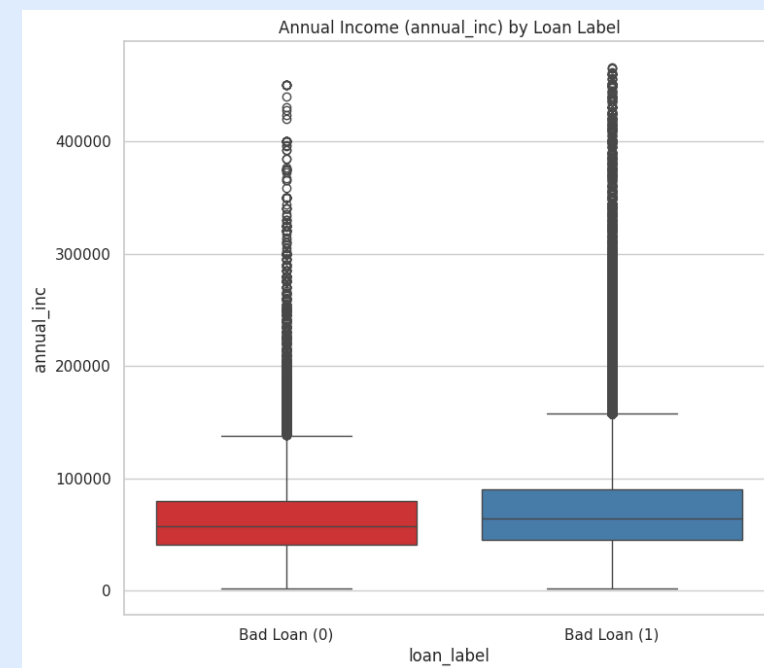
2. EXPLORATORY DATA ANALYSIS (EDA)



- Bad loans tend to have **higher median interest rates** than good loans.
- Higher interest rates reflect **higher underlying credit risk**.
- Interest rate is a **strong discriminating feature** between loan outcomes.



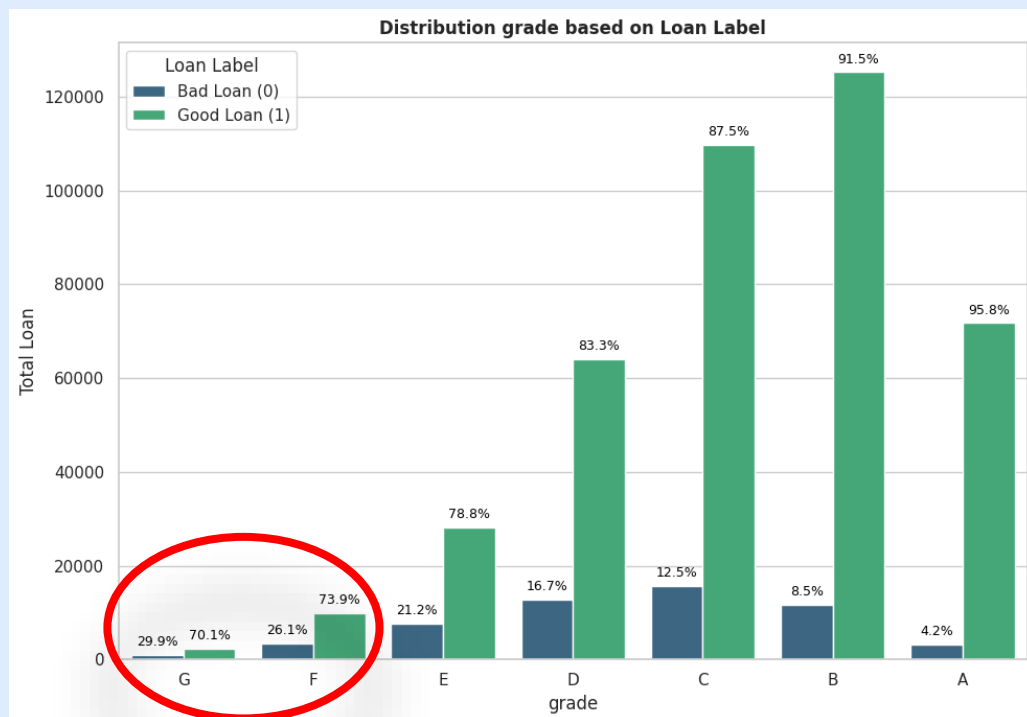
- Bad loans generally show **higher DTI distributions**.
- Borrowers with high DTI face **greater repayment pressure**.
- DTI is a **critical indicator of default risk**.



- Good loans are associated with **higher median annual income**.
- Bad loan borrowers show **lower income levels** with higher variability.
- Income alone is not sufficient; it must be analyzed alongside DTI.

Bivariate Analysis

2. EXPLORATORY DATA ANALYSIS (EDA)



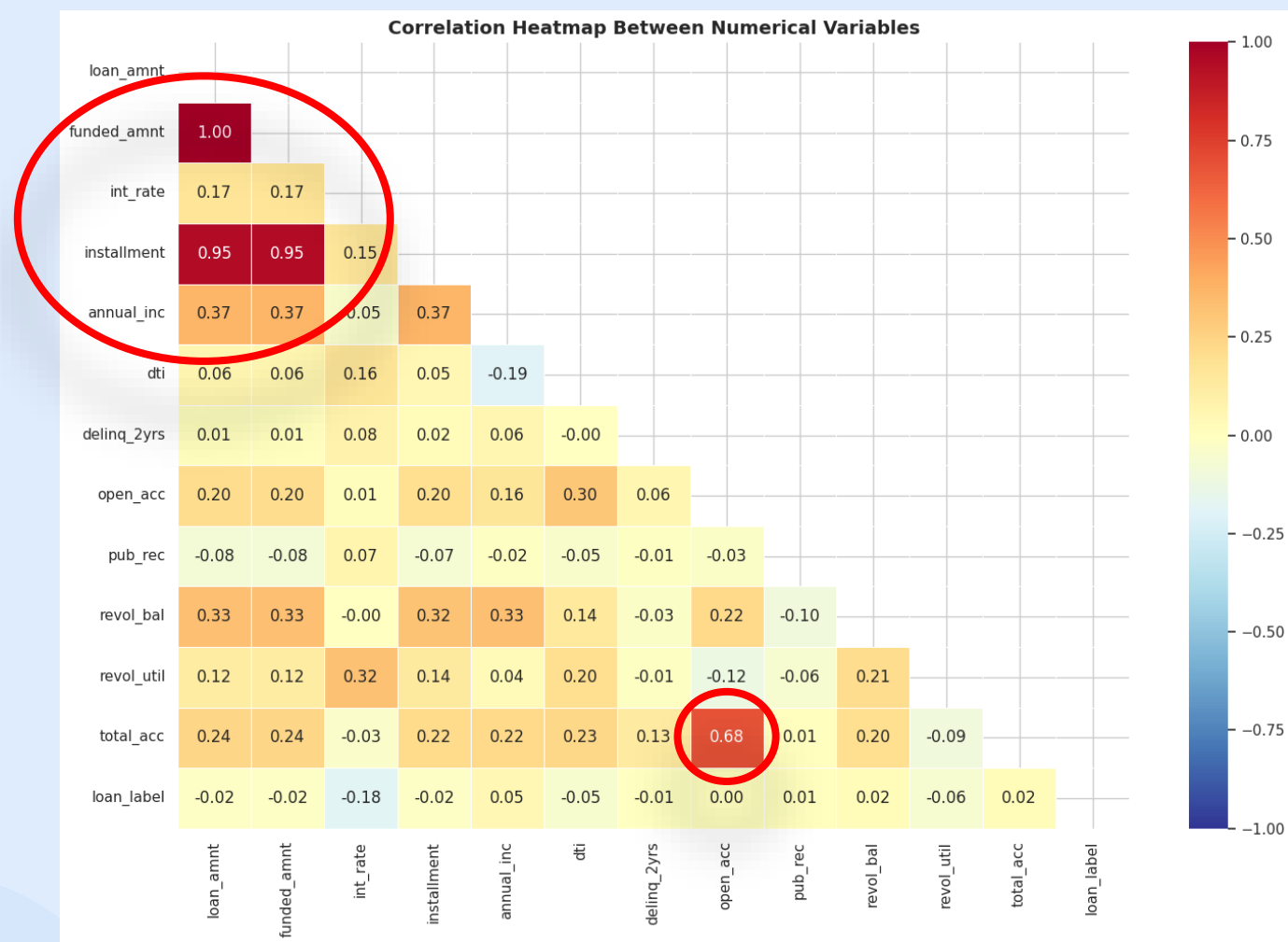
- Higher credit grades (**A–B**) show a **very high proportion of good loans**, exceeding 90% repayment rate.
- Lower credit grades (**E–G**) exhibit a **significantly higher proportion of bad loans**.
- Default rates increase **consistently as credit grade deteriorates**.



- Borrowers with **Mortgage** status show a **high proportion of good loans** (**≈89.5%**).
- **Home owners (OWN)** also demonstrate strong repayment performance compared to renters.
- **Renters (RENT)** contribute the largest volume of bad loans in absolute terms.

Bivariate Analysis

2. EXPLORATORY DATA ANALYSIS (EDA)



Strong Positive Correlations

- **Loan amount and funded amount** show a **perfect correlation** (≈ 1.0).
- **Loan amount and installment** show a **very strong positive correlation** (≈ 0.95).
- This indicates clear mathematical relationships driven by loan structure.

Moderate Financial Relationships

- **Annual income** moderately correlates with **loan amount and installment** (≈ 0.37).
- Borrowers with higher income tend to take **larger loans and higher installments**, but income alone does not strongly drive borrowing behavior.

Credit Behavior Indicators

- **Total accounts and open accounts** show a **strong positive correlation** (≈ 0.68).
- **Revolving balance** is moderately correlated with **loan amount and income**, reflecting accumulated credit usage.
- **Revolving utilization** shows only weak correlation with most variables, indicating independent risk behavior.

Correlation Analysis

Feature Engineering

— *Business Logic Enrichment*



2. FEATURE ENGINEERING

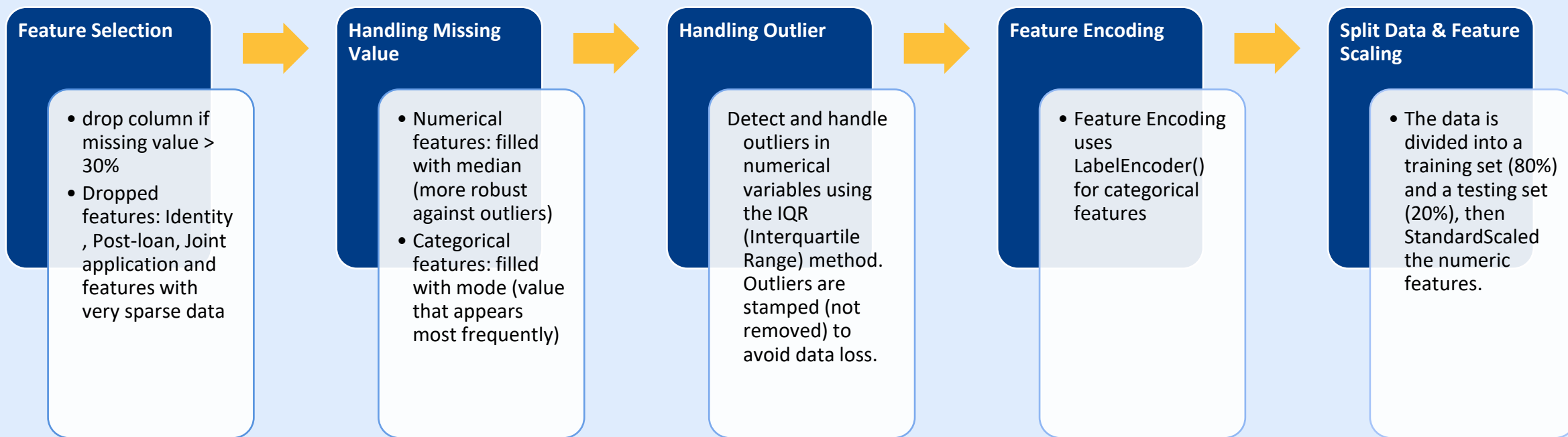
Feature Extraction by adding 3 features to explain the history of late loan payments

No	Feature		Syntax
1.	Historical Delay Flag (Binary) : Customer EVER paid late in the last 2 years	has_history_delinq	<pre>if 'delinq_2yrs' in df.columns: df['has_history_delinq'] = np.where(df['delinq_2yrs'] > 0, 1, 0)</pre>
2.	Derogatory Public Records (Black Records / Bankruptcy)	has_public_record	<pre>if 'pub_rec' in df.columns: df['has_public_record'] = np.where(df['pub_rec'] > 0, 1, 0)</pre>
3.	Currently Delinquent (Status in Arrears at the Time of Filing)	is_currently_delinq	<pre>if 'acc_now_delinq' in df.columns: df['is_currently_delinq'] = np.where(df['acc_now_delinq'] > 0, 1, 0)</pre>

Data Preparation



4. DATA PREPARATION



MODELING STRATEGY

— *Not All Models Are Equal*



5. DATA MODELING

Model Algoritm	Accuracy	Precision	Recall	F1-Score	F2-Score	ROC-AUC
Logistic Regression	0.6303	0.9273	0.6299	0.7502	0.6731	0.6826
Random Forest	0.7428	0.9126	0.7832	0.843	0.8061	0.685
Decision Tree	0.5954	0.9288	0.5858	0.7185	0.6325	0.6688
Gradient Boosting	0.8814	0.8815	0.9997	0.9369	0.9736	0.6978
XGBoost	0.6253	0.9325	0.6197	0.7446	0.6642	0.6972

- Gradient Boosting achieves the best overall performance, with:
 - Highest Accuracy (0.8814)
 - Near-perfect Recall (0.9997)
 - Strong F1-Score (0.9369) and F2-Score (0.9736)
- Random Forest shows a good balance between precision and recall, but underperforms compared to Gradient Boosting.
- Logistic Regression and Decision Tree exhibit lower recall, increasing the risk of misclassifying bad loans.
- XGBoost delivers high precision but relatively lower recall in this setup.

5. DATA MODELING

Benchmarking Model Results (*After RandomizedSearchCV Tuning*)

Metric	Logistic Regression		Decision Tree		Random Forest		Gradient Boosting		XGBoost	
	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
Accuracy	0.6282	0.63	0.603	0.6047	0.6428	0.6329	0.8815	0.8813	0.8814	0.8813
Precision	0.9271	0.9273	0.9277	0.9283	0.938	0.9282	0.8815	0.8814	0.8814	0.8813
Recall	0.6274	0.6296	0.5959	0.5976	0.6368	0.6324	1	0.9999	1	1
F1 Score	0.7484	0.75	0.7257	0.7271	0.7586	0.7523	0.937	0.9369	0.9369	0.9369
F2 Score	0.6708	0.6728	0.6418	0.6434	0.6805	0.6754	0.9738	0.9737	0.9738	0.9738
ROC-AUC	0.6798	0.6826	0.6716	0.6737	0.7251	0.6881	0.7028	0.6948	0.7254	0.6994

- Gradient Boosting and XGBoost deliver the best overall performance across almost all metrics.
- Both models achieve:
 - High Accuracy (~88%)
 - Perfect or near-perfect Recall (~1.00)
 - Strong F1-Score (~0.937) and F2-Score (~0.974)
- Performance between train and test sets is highly consistent, indicating good generalization.

Model Evaluation

— *Business Lens*



6. EVALUATION

Model Evaluation (Focus on Bad Loan Detection)

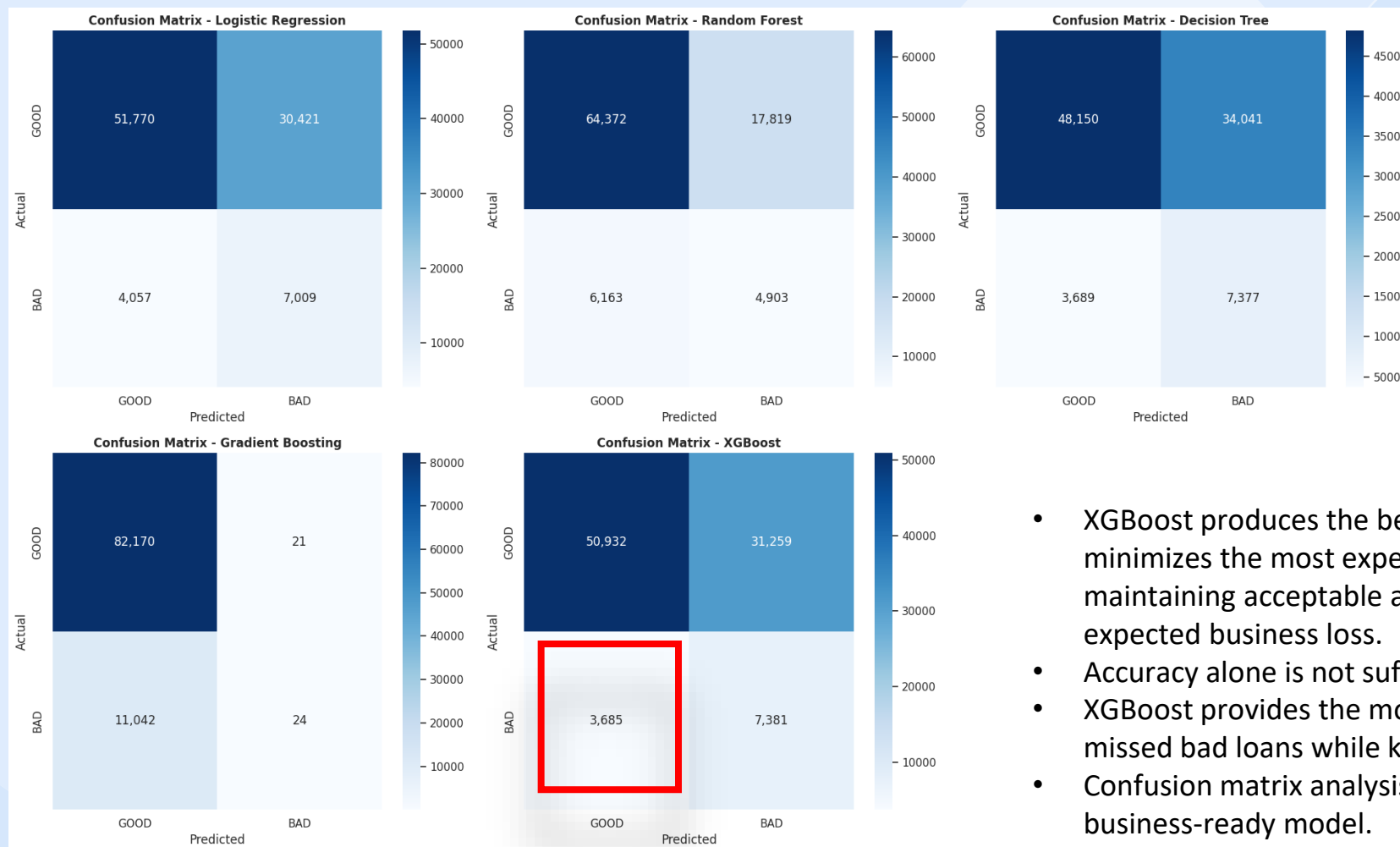
Dataset	Recall (Bad Loan)		F2-Score		Accuracy	
Model	Test	Train	Test	Train	Test	Train
Decision Tree	0.666637	0.713934	0.430487	0.460521	0.595419	0.605611
Gradient Boosting	0.002169	0.004383	0.002708	0.005471	0.881371	0.881781
Logistic Regression	0.633382	0.633556	0.428979	0.428154	0.63029	0.628384
Random Forest	0.443069	0.780599	0.365972	0.631968	0.74284	0.808384
XGBoost	0.666998	0.686147	0.445153	0.456751	0.625294	0.627519

- Decision Tree and XGBoost achieve the highest recall for bad loans (~0.67) on the test set.
- XGBoost delivers the highest F2-score, prioritizing recall over precision.
- Gradient Boosting and Random Forest achieve higher accuracy, but at the cost of missing most bad loans.
- XGBoost is the most suitable model for bad loan detection, balancing recall, F2-score, and acceptable accuracy.

- Missing a bad loan is more costly than rejecting a good loan
- Model selection should prioritize:
 - Recall (Bad Loan)
 - F2-score
 - Business risk exposure
- XGBoost aligns best with real-world credit risk objectives

6. EVALUATION

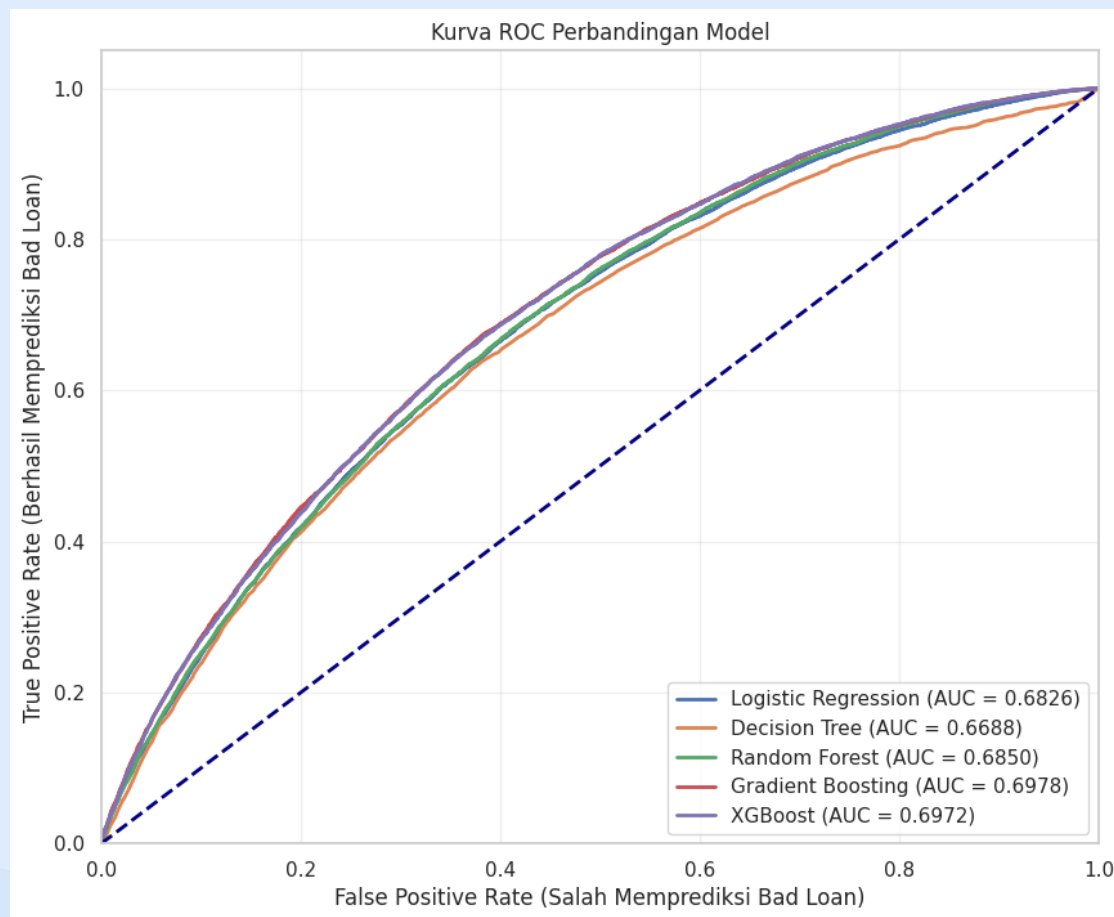
Confusion Matrix



- XGBoost produces the best confusion matrix because it minimizes the most expensive error—false negatives—while maintaining acceptable approval accuracy, resulting in optimal expected business loss.
- Accuracy alone is not sufficient for credit risk evaluation.
- XGBoost provides the most effective trade-off, minimizing missed bad loans while keeping approval rates reasonable.
- Confusion matrix analysis confirms XGBoost as the most business-ready model.

6. EVALUATION

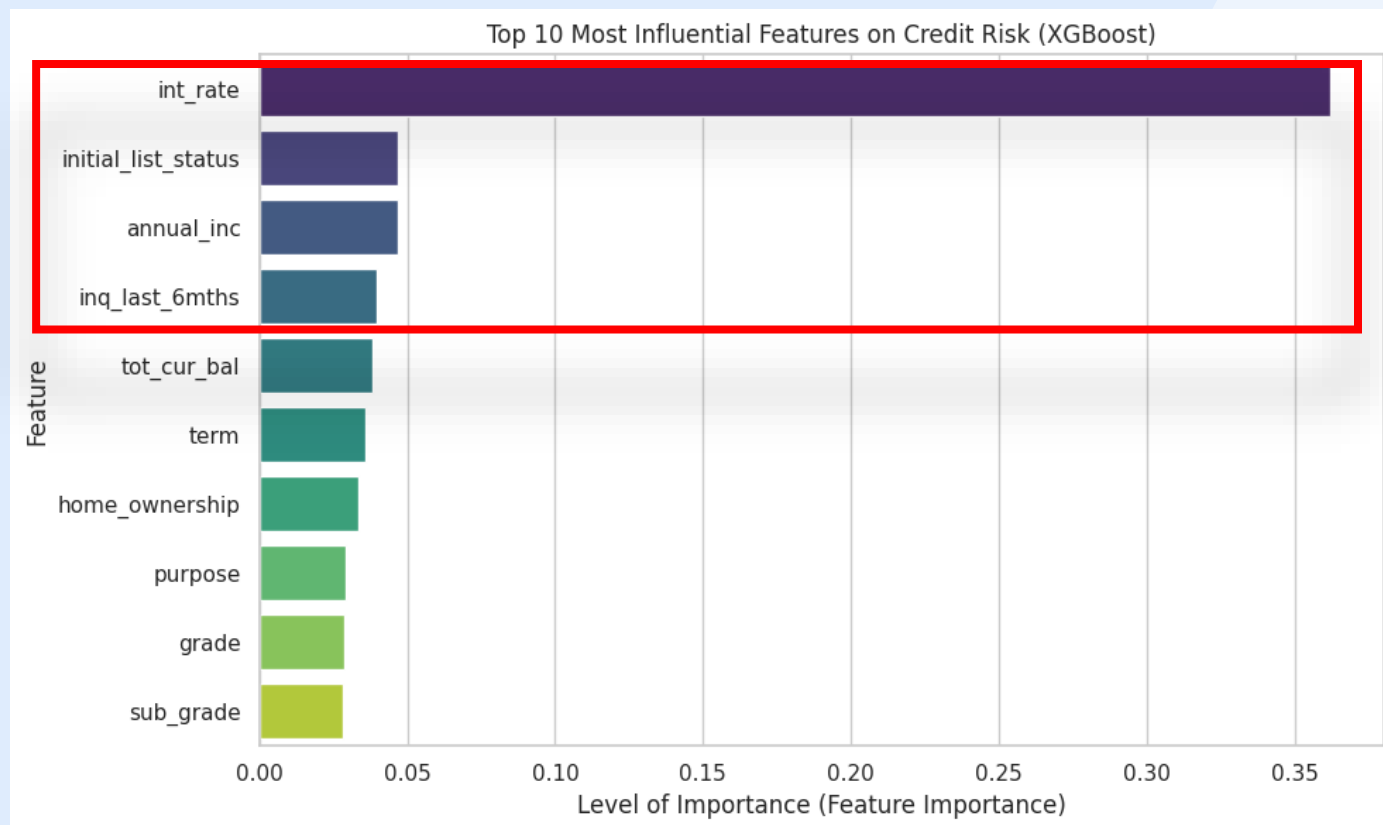
ROC (Receiver Operating Characteristic) Curve



- All models perform **better than random guessing**, as indicated by ROC curves above the diagonal baseline.
- **Gradient Boosting and XGBoost** provide the **strongest overall discrimination ability**.
- Boosting-based models demonstrate **more consistent performance across different thresholds**.
- ROC analysis supports the use of **ensemble boosting models** for credit risk prediction.
- Final model choice should combine ROC-AUC with **business-focused metrics such as recall and confusion matrix impact**.

6. EVALUATION

Feature Importance

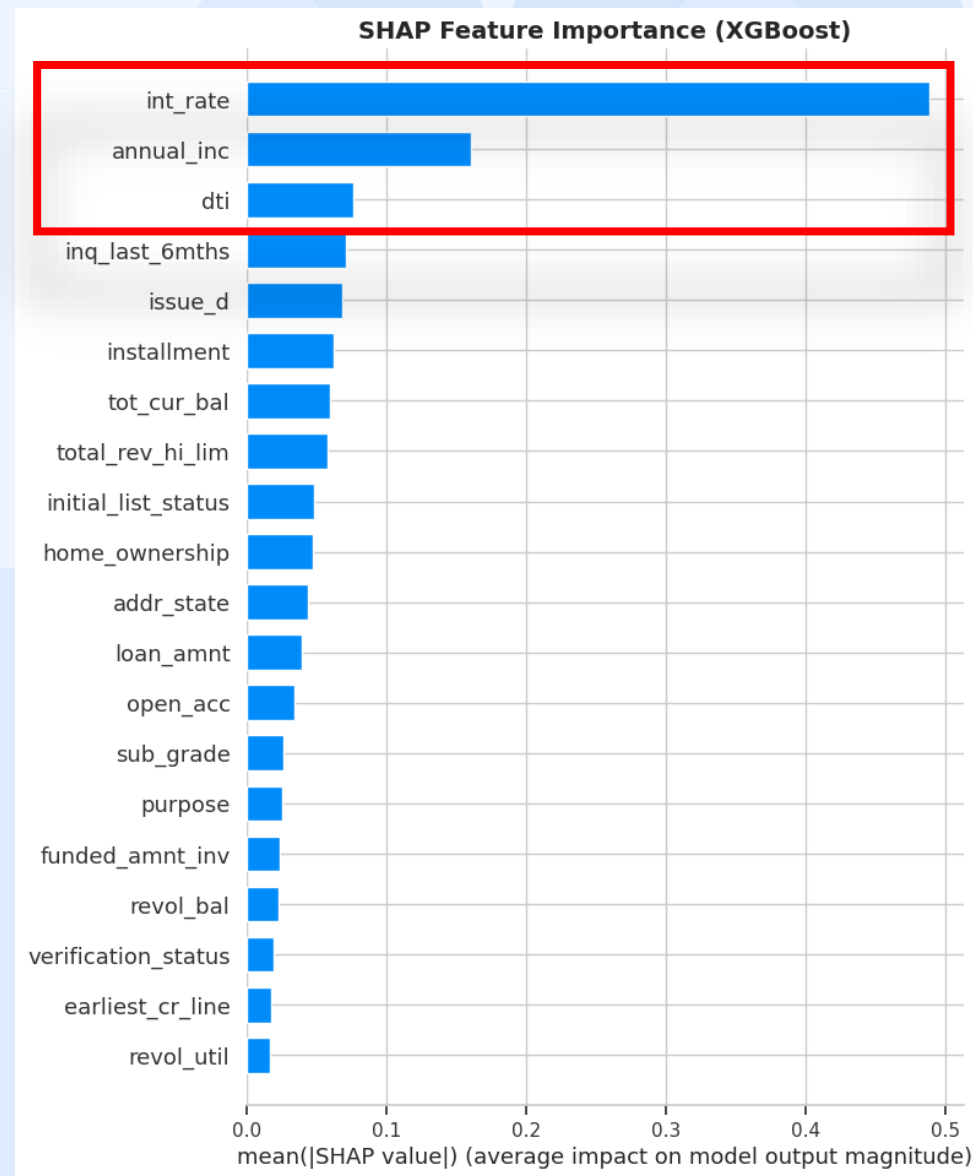


- **Interest Rate (int_rate)** is the most influential feature by a large margin.
- Financial and credit-related variables such as: **annual_inc, tot_cur_bal, inq_last_6mths** play a significant role in risk prediction.
- XGBoost prioritizes **pricing and borrower affordability** over single demographic factors.
- Credit inquiry behavior (inq_last_6mths) highlights **recent credit stress**.
- Loan grading variables reinforce the **validity of existing credit scoring frameworks**.
- **Interest rate is the strongest proxy for credit risk**, reflecting cumulative risk assessment.
- Combining financial capacity, credit behavior, and loan structure enables robust risk prediction.
- The model is **interpretable and aligned with real-world credit decision logic**.

7. EXPLAINABILITY — TRUST & REGULATION

SHAP Feature Importance

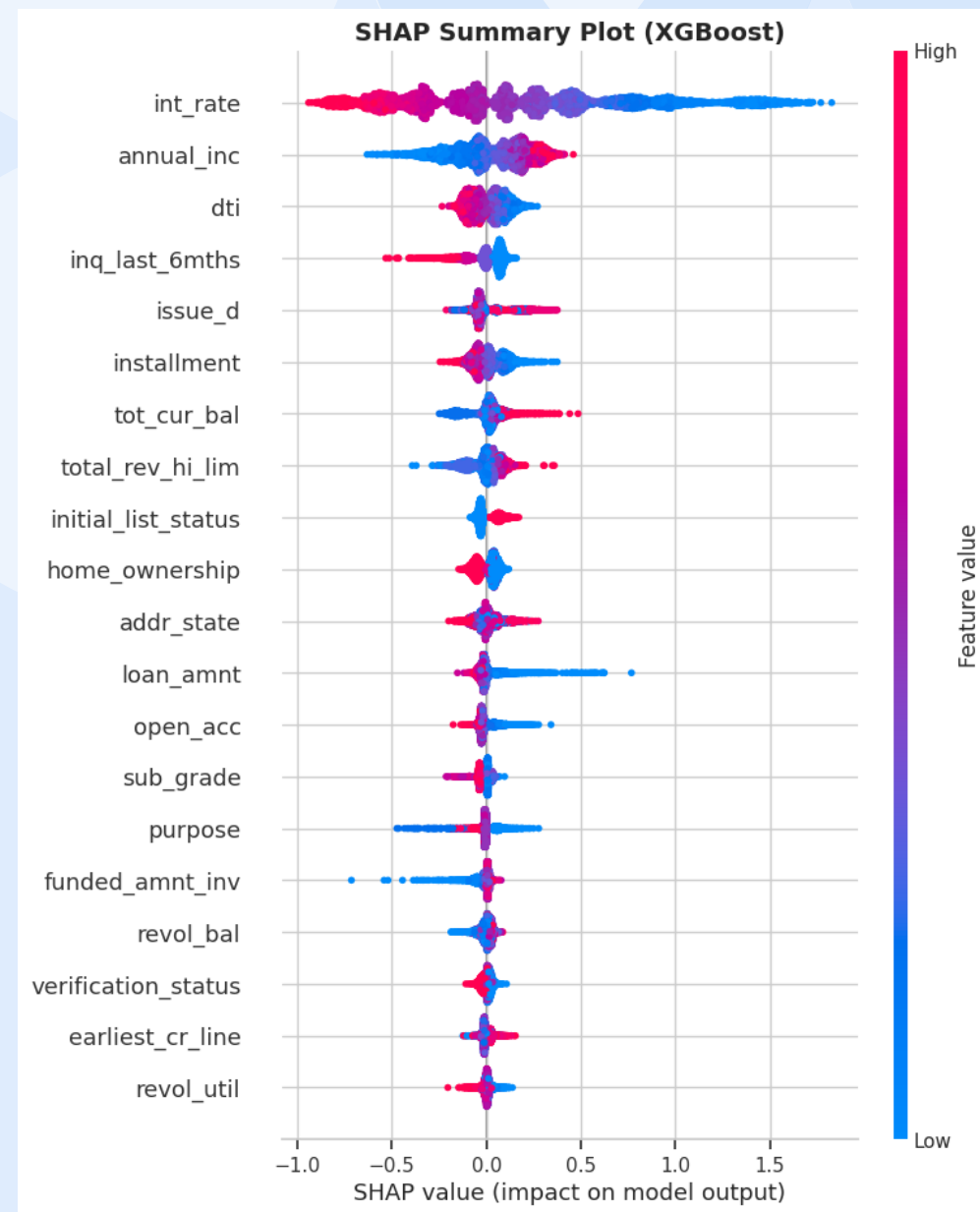
- **Interest Rate (int_rate)** is the most influential feature by a significant margin.
- Financial capacity indicators such as: **annual_inc**, **dti**, **installment** strongly affect credit risk prediction.
- Credit behavior variables including: **inq_last_6mths**, **tot_cur_bal**, **revol_bal** capture recent financial stress and utilization patterns.
- SHAP confirms that risk pricing, affordability, and recent credit behavior dominate model decisions.
- The model relies less on static demographics and more on dynamic financial risk indicators.
- This improves fairness, transparency, and regulatory trust.



7. EXPLAINABILITY — TRUST & REGULATION

SHAP Summary Plot (XGBoost)

- **Interest Rate (int_rate):** High interest rates strongly push predictions toward bad loan. Low interest rates reduce risk significantly. Confirms interest rate as the strongest risk signal.
- **Annual Income (annual_inc) :** Higher income values push predictions toward lower risk. Lower income contributes to higher default risk.
- **Debt-to-Income (dti) :** High DTI consistently increases bad loan probability. Confirms affordability pressure as a critical risk driver.
- **Recent Credit Inquiries (inq_last_6mths) :** High inquiry counts are associated with higher credit risk. Captures short-term financial stress behavior.
- The model decisions are driven primarily by pricing, affordability, and recent credit behavior.
- SHAP Summary confirms directional consistency and explainability of XGBoost predictions.
- The model aligns well with real-world credit risk intuition.



BUSINESS IMPACT

Model Output (Confusion Matrix – XGBoost)

TN (Good Loan Approved) 50,932	FP (Good Loan Rejected) 31,259
FN (Bad Loan Missed): 3,685	TP (Bad Loan Detected) 7,381

Simulation Assumptions

- Average loan value: IDR 100 million per loan
- Cost of False Negative (Bad loan approved): 100% loss
- Cost of False Positive (Good loan rejected): 10% opportunity loss

Component	Calculate	Value
Risk Prevented (TP)	7,381 loans × IDR 100M	IDR 738.1B
Residual Loss (FN)	3,685 loans × IDR 100M	IDR 368.5B
Opportunity Loss (FP)	31,259 loans × IDR 100M × 10%	IDR 312.6B

Key Insight

- XGBoost successfully blocks 66% of potential bad loans, significantly reducing financial risk.
- Although some good loans are rejected, the risk prevented outweighs opportunity loss.
- The model provides the best trade-off between risk mitigation and business growth.

Conclusion



7. CONCLUSION

- Developed an **end-to-end credit risk modeling pipeline** from EDA to explainable AI.
- **Key risk drivers** identified: *interest rate, DTI, income level, credit grade, and recent credit behavior.*
- **Ensemble models outperform traditional methods**, confirming non-linear risk patterns.
- **XGBoost selected as the final model**, delivering:
 - High **recall for bad loan detection**
 - Strong **F2-score** aligned with risk mitigation
 - Stable and competitive **ROC-AUC**
- **SHAP explainability** confirms decisions are **transparent, logical, and regulation-ready.**
- XGBoost retains more than **66% of credit risk** by correctly identifying the majority of bad loans, as measured by bad-loan recall.

Thank You

KUNTUR JALASSUAD

+62-817-7087-3794

JALASSUAD.K@GMAIL.COM

[HTTPS://WWW.LINKEDIN.COM/IN/KUNTUR-JALASSUAD/](https://www.linkedin.com/in/kuntur-jalassuad/)



X

id/x partners

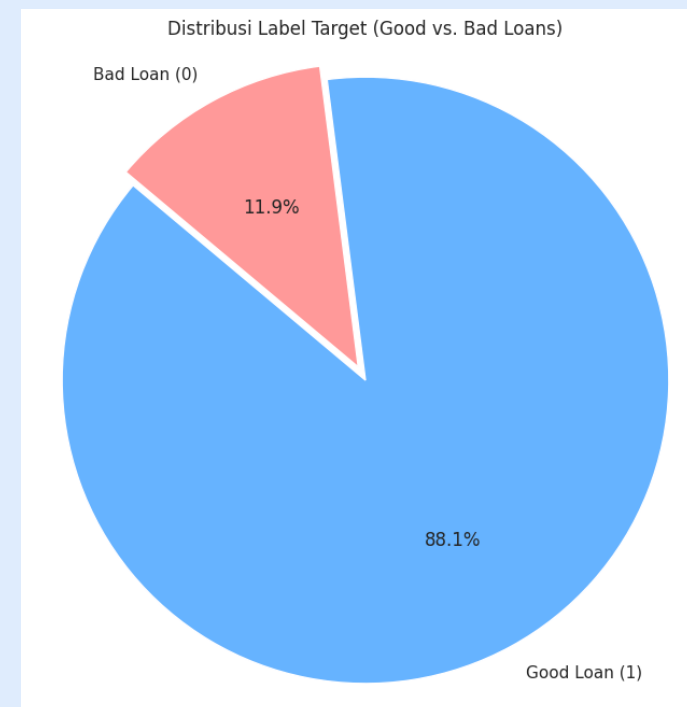
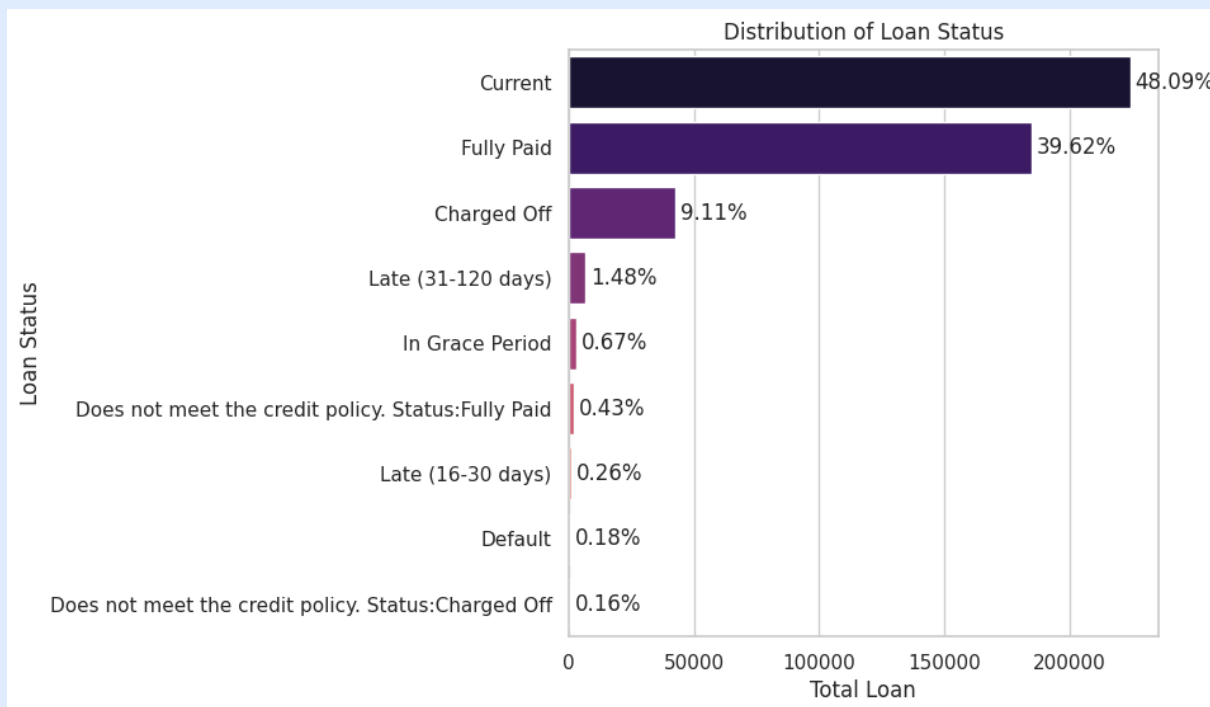
APPENDIX

1. DATA UNDERSTANDING

Dataset : loan_data_2007_2014.csv		
File Size	228	MB
Data Shape	466285	Rows
	75	Feature
Data Type	46	float64
	7	int64
	22	object

Missing Values : 40 out of 75 columns		
Missing Value (%)	Sum of Feature	%
100%	17	23%
100% < x > 30%	5	7%
< 30%	18	24%
0%	35	47%

1. DATA UNDERSTANDING



Categories		Loan Status Distribution	Count Value
Good	Current		224,226
Good	Fully Paid		184,739
Good	Does not meet the credit policy. Status:Fully Paid		1,988
Total (Good Status)			410,953 (88.14%)

Categories	Loan Status Distribution	Count Value
Bad	Charged Off	42,475
Bad	Late (31-120 days)	6,900
Bad	In Grace Period	3,146
Bad	Late (16-30 days)	1,218
Bad	Default	832
Bad	Does not meet the credit policy. Status:Charged Off	761
Total (Bad Status)		55,332 (11.86%)

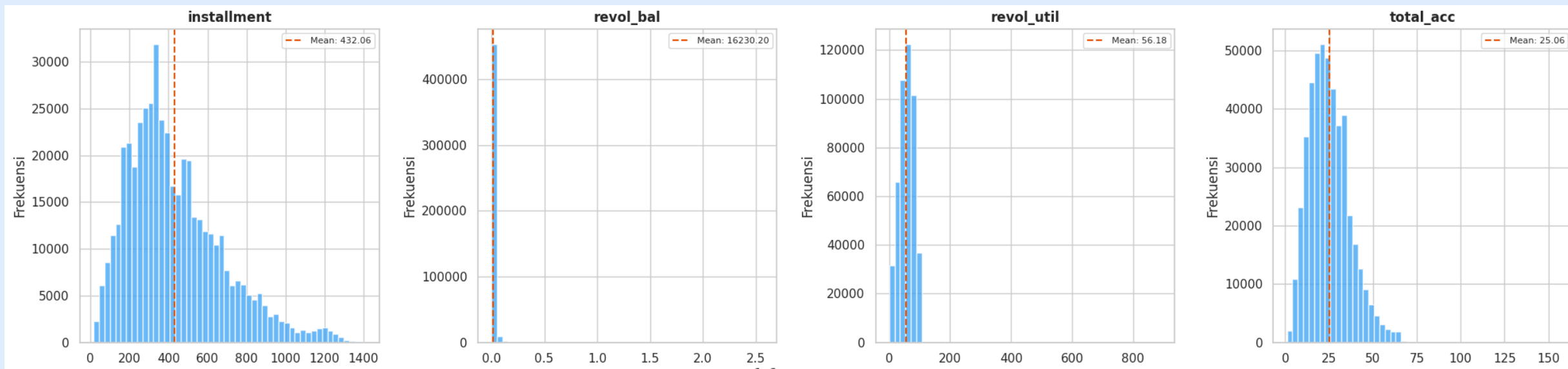
STATISTIC NUMERICAL

	count	mean	std	min	25%	50%	75%	max
Unnamed: 0	466285.0	2.331420e+05	1.346050e+05	0.00	1.165710e+05	2.331420e+05	3.497130e+05	4.662840e+05
id	466285.0	1.307973e+07	1.089371e+07	54734.00	3.639987e+06	1.010790e+07	2.073121e+07	3.809811e+07
member_id	466285.0	1.459766e+07	1.168237e+07	70473.00	4.379705e+06	1.194108e+07	2.300154e+07	4.086083e+07
loan_amnt	466285.0	1.431728e+04	8.286509e+03	500.00	8.000000e+03	1.200000e+04	2.000000e+04	3.500000e+04
funded_amnt	466285.0	1.429180e+04	8.274371e+03	500.00	8.000000e+03	1.200000e+04	2.000000e+04	3.500000e+04
funded_amnt_inv	466285.0	1.422233e+04	8.297638e+03	0.00	8.000000e+03	1.200000e+04	1.995000e+04	3.500000e+04
int_rate	466285.0	1.382924e+01	4.357587e+00	5.42	1.099000e+01	1.366000e+01	1.649000e+01	2.606000e+01
installment	466285.0	4.320612e+02	2.434855e+02	15.67	2.566900e+02	3.798900e+02	5.665800e+02	1.409990e+03
annual_inc	466281.0	7.327738e+04	5.496357e+04	1896.00	4.500000e+04	6.300000e+04	8.896000e+04	7.500000e+06
dti	466285.0	1.721876e+01	7.851121e+00	0.00	1.136000e+01	1.687000e+01	2.278000e+01	3.999000e+01
delinq_2yrs	466256.0	2.846784e-01	7.973651e-01	0.00	0.000000e+00	0.000000e+00	0.000000e+00	2.900000e+01
inq_last_6mths	466256.0	8.047446e-01	1.091598e+00	0.00	0.000000e+00	0.000000e+00	1.000000e+00	3.300000e+01
mths_since_last_delinq	215934.0	3.410443e+01	2.177849e+01	0.00	1.600000e+01	3.100000e+01	4.900000e+01	1.880000e+02
mths_since_last_record	62638.0	7.430601e+01	3.035765e+01	0.00	5.300000e+01	7.600000e+01	1.020000e+02	1.290000e+02
open_acc	466256.0	1.118707e+01	4.987526e+00	0.00	8.000000e+00	1.000000e+01	1.400000e+01	8.400000e+01
pub_rec	466256.0	1.605642e-01	5.108626e-01	0.00	0.000000e+00	0.000000e+00	0.000000e+00	6.300000e+01
revol_bal	466285.0	1.623020e+04	2.067625e+04	0.00	6.413000e+03	1.176400e+04	2.033300e+04	2.568995e+06
revol_util	465945.0	5.617695e+01	2.373263e+01	0.00	3.920000e+01	5.760000e+01	7.470000e+01	8.923000e+02
total_acc	466256.0	2.506443e+01	1.160014e+01	1.00	1.700000e+01	2.300000e+01	3.200000e+01	1.560000e+02
out_prncp	466285.0	4.410062e+03	6.355079e+03	0.00	0.000000e+00	4.414700e+02	7.341650e+03	3.216038e+04
out_prncp_inv	466285.0	4.408452e+03	6.353198e+03	0.00	0.000000e+00	4.413800e+02	7.338390e+03	3.216038e+04
total_pymnt	466285.0	1.154069e+04	8.265627e+03	0.00	5.552125e+03	9.419251e+03	1.530816e+04	5.777758e+04
total_pymnt_inv	466285.0	1.146989e+04	8.254158e+03	0.00	5.499250e+03	9.355430e+03	1.523131e+04	5.777758e+04
total_rec_prncp	466285.0	8.866015e+03	7.031688e+03	0.00	3.708560e+03	6.817760e+03	1.200000e+04	3.500003e+04
total_rec_int	466285.0	2.588677e+03	2.483810e+03	0.00	9.572800e+02	1.818880e+03	3.304530e+03	2.420562e+04
total_rec_late_fee	466285.0	6.501292e-01	5.265730e+00	0.00	0.000000e+00	0.000000e+00	0.000000e+00	3.586800e+02

	count	mean	std	min	25%	50%	75%	max
recoveries	466285.0	8.534421e+01	5.522161e+02	0.00	0.000000e+00	0.000000e+00	0.000000e+00	3.352027e+04
collection_recovery_fee	466285.0	8.961534e+00	8.549144e+01	0.00	0.000000e+00	0.000000e+00	0.000000e+00	7.002190e+03
last_pymnt_amnt	466285.0	3.123914e+03	5.554737e+03	0.00	3.126200e+02	5.459600e+02	3.187510e+03	3.623444e+04
collections_12_mths_ex_med	466140.0	9.085253e-03	1.086484e-01	0.00	0.000000e+00	0.000000e+00	0.000000e+00	2.000000e+01
mths_since_last_major_derog	98974.0	4.285255e+01	2.166259e+01	0.00	2.600000e+01	4.200000e+01	5.900000e+01	1.880000e+02
policy_code	466285.0	1.000000e+00	0.000000e+00	1.00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
annual_inc_joint	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
dti_joint	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
verification_status_joint	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
acc_now_delinq	466256.0	4.002093e-03	6.863680e-02	0.00	0.000000e+00	0.000000e+00	0.000000e+00	5.000000e+00
tot_coll_amt	396009.0	1.919135e+02	1.463021e+04	0.00	0.000000e+00	0.000000e+00	0.000000e+00	9.152545e+06
tot_cur_bal	396009.0	1.388017e+05	1.521147e+05	0.00	2.861800e+04	8.153900e+04	2.089530e+05	8.000078e+06
open_acc_6m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
open_il_6m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
open_il_12m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
open_il_24m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
mths_since_rcnt_il	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
total_bal_il	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
il_util	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
open_rv_12m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
open_rv_24m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
max_bal_bc	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
all_util	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
total_rev_hi_lim	396009.0	3.037909e+04	3.724713e+04	0.00	1.350000e+04	2.280000e+04	3.790000e+04	9.999999e+06
inq_fi	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
total_cu_tl	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
inq_last_12m	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN

2. EXPLORATORY DATA ANALYSIS (EDA)

Univariate Analysis



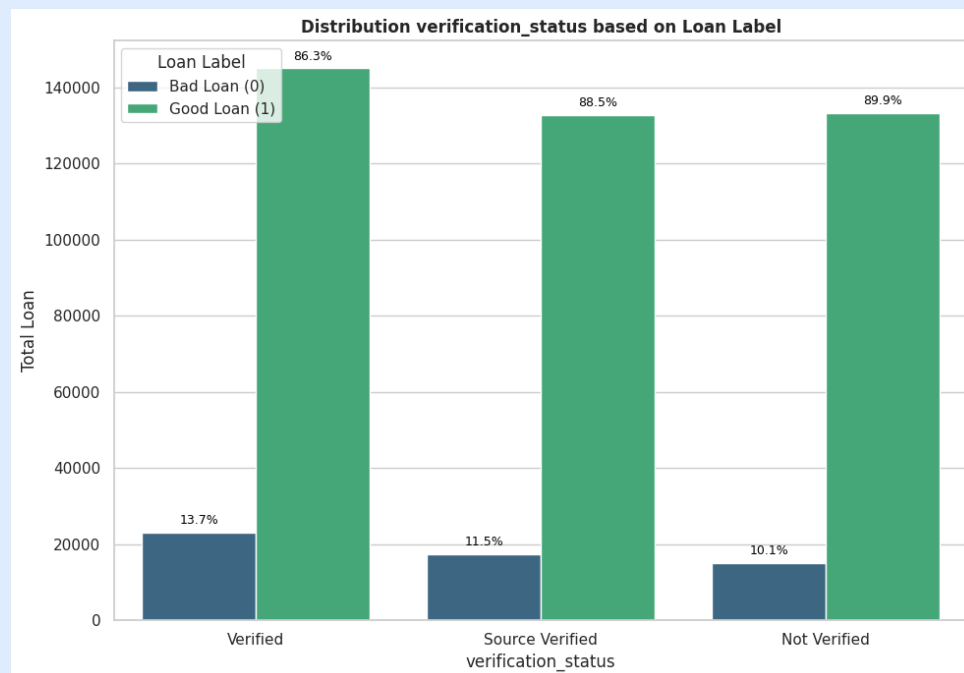
Installment amounts **are right-skewed**, driven by loan amount and loan term.

Revolving balance is **heavily right-skewed**, with most borrowers carrying small balances and a few extreme cases.

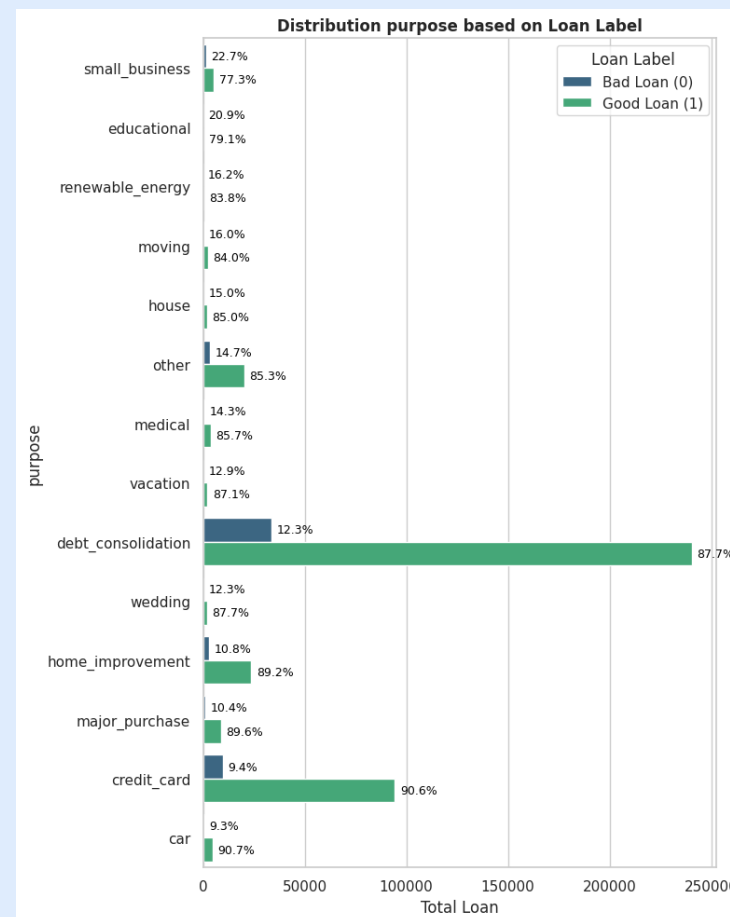
Most borrowers maintain **below 50% revolving utilization**, but a right tail approaches full utilization.

The distribution is unimodal, with most borrowers holding **10–40 credit accounts**.

2. EXPLORATORY DATA ANALYSIS (EDA)



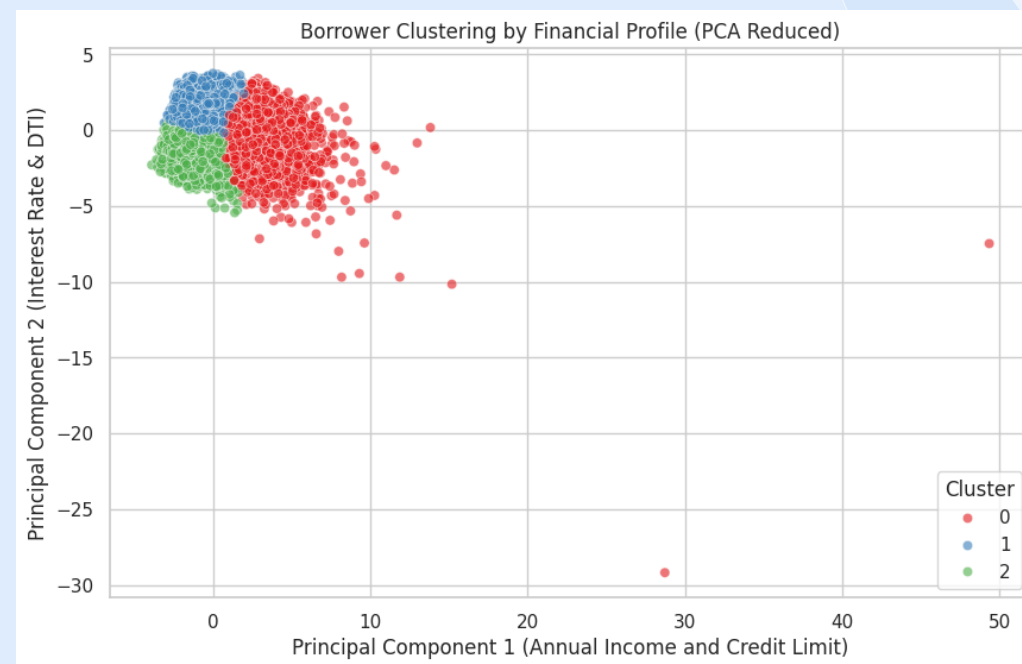
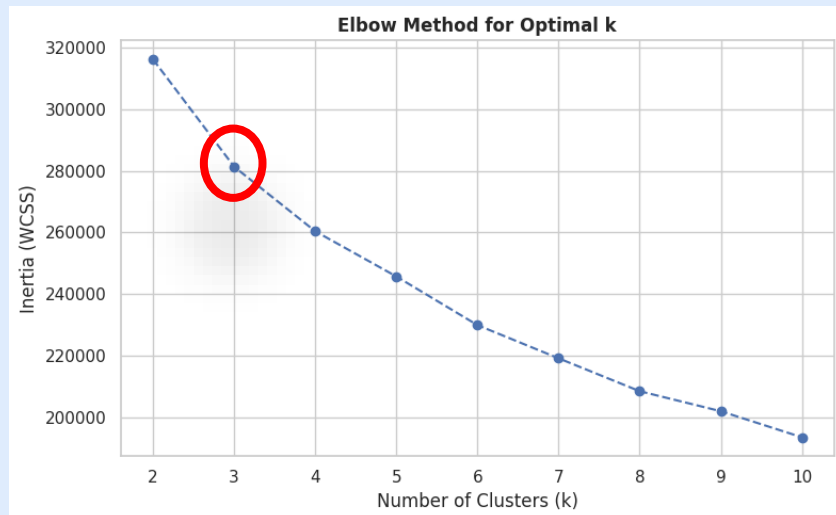
- All verification categories show a **high proportion of good loans**, exceeding **85%**.
- **Not Verified** loans surprisingly exhibit the **highest good loan rate (~89.9%)**.
- **Verified** loans display a slightly higher proportion of bad loans compared to other categories.



- **Debt consolidation** dominates the portfolio and shows a **high good loan rate (~87.7%)**.
- **Small business** loans have the **highest bad loan proportion (~22.7%)** among all purposes.
- Consumer-related purposes such as **car**, **credit card**, and **major purchase** exhibit **strong repayment performance (>89%)**.

Bivariate Analysis

2. EXPLORATORY DATA ANALYSIS (EDA)



Cluster 0 — High Income, High Leverage Borrower	Cluster 1 — Stressed Borrower with High Utilization	Cluster 2 — Low Risk, Financially Disciplined Borrowers
This borrower has a stable income, but is very active in debt and has high leverage	This segment includes borrowers who have used almost their entire credit limit. They have the least financial breathing room.	This segment of borrowers is not aggressive in debt. Manages credit very well. Income is not the highest, but credit quality is the strongest.
Key Characteristics:		
1. Highest loan_amount ≈ 25.4k 2. Highest annual_inc ≈ 110k 3. Large installments → heavy loan burden 4. Revol_utility 61.8% (quite high) 5. Total_acc 30.9% (very many credit accounts) 6. Loan_label 0.879 (good, but not the best)	1. Lowest annual_inc ≈ 56k 2. Highest dti ≈ 20.45 3. Highest revol_util ≈ 69.6% 4. Highest int_rate (15.6%) 5. Loan_label 0.848 (lowest)	1. Lowest loan_amnt ≈ 10.1k 2. Lowest int_rate ≈ 11.1% 3. Lowest dti ≈ 13.0 4. Very low revol_util ≈ 36.2% 5. Highest loan_label ≈ 0.921

Clustering

4. DATA PREPARATION

- Feature Selection

Dropped features:

- Identity features: id, member_id, url, desc, title, emp_title, zip_code
- Status features already used as labels: loan_label
- Post-loan features (data only available after a loan is disbursed): out_prncp, total_pymnt, recoveries, etc.
- Joint application features (too many missing): annual_inc_joint, dti_joint, etc.
- Features with very sparse data: open_acc_6m, open_il_6m, etc.

Features before selection	Features After selection
79	35

Features used:

```
['loan_amnt', 'funded_amnt', 'funded_amnt_inv', 'term', 'int_rate', 'installment', 'grade', 'sub_grade', 'emp_length', 'home_ownership', 'annual_inc', 'verification_status', 'issue_d', 'loan_status', 'purpose', 'addr_state', 'dti', 'delinq_2yrs', 'earliest_cr_line', 'inq_last_6mths', 'open_acc', 'pub_rec', 'revol_bal', 'revol_util', 'total_acc', 'initial_list_status', 'collections_12_mths_ex_med', 'acc_now_delinq', 'tot_coll_amt', 'tot_cur_bal', 'total_rev_hi_lim', 'loan_label', 'has_history_delinq', 'has_public_record', 'is_currently_delinq']
```

```
Data columns (total 35 columns):
#      Column                                     Non-Null Count  Dtype
---  -
0      loan_amnt                                     466285 non-null  int64
1      funded_amnt                                   466285 non-null  int64
2      funded_amnt_inv                               466285 non-null  float64
3      term                                           466285 non-null  object
4      int_rate                                       466285 non-null  float64
5      installment                                   466285 non-null  float64
6      grade                                           466285 non-null  object
7      sub_grade                                      466285 non-null  object
8      emp_length                                    445277 non-null  object
9      home_ownership                               466285 non-null  object
10     annual_inc                                    466281 non-null  float64
11     verification_status                           466285 non-null  object
12     issue_d                                       466285 non-null  object
13     loan_status                                   466285 non-null  object
14     purpose                                       466285 non-null  object
15     addr_state                                   466285 non-null  object
16     dti                                           466285 non-null  float64
17     delinq_2yrs                                  466256 non-null  float64
18     earliest_cr_line                             466256 non-null  object
19     inq_last_6mths                               466256 non-null  float64
20     open_acc                                     466256 non-null  float64
21     pub_rec                                      466256 non-null  float64
22     revol_bal                                    466285 non-null  int64
23     revol_util                                   465945 non-null  float64
24     total_acc                                    466256 non-null  float64
25     initial_list_status                           466285 non-null  object
26     collections_12_mths_ex_med                   466140 non-null  float64
27     acc_now_delinq                               466256 non-null  float64
28     tot_coll_amt                                 396009 non-null  float64
29     tot_cur_bal                                  396009 non-null  float64
30     total_rev_hi_lim                             396009 non-null  float64
31     loan_label                                   466285 non-null  int64
32     has_history_delinq                           466285 non-null  int64
33     has_public_record                            466285 non-null  int64
34     is_currently_delinq                           466285 non-null  int64
dtypes: float64(16), int64(7), object(12)
memory usage: 124.5+ MB
```

4. DATA PREPARATION

- Handling Missing Value

```
1 # Imputasi nilai yang hilang
2 df_clean[num_cols] = df_clean[num_cols].fillna(df_clean[num_cols].median())
3 df_clean[cat_cols] = df_clean[cat_cols].fillna(df_clean[cat_cols].mode().iloc[0])
4
5 print("Missing values after imputation:\n", df_clean.isnull().sum().max())
```

Missing values after imputation:
0

- Handling Outlier

```
1 outlier_cols = ['annual_inc', 'dti', 'revol_bal', 'revol_util', 'delinq_2yrs']
2 available_outlier = [c for c in outlier_cols if c in df_clean.columns]
3
4 print("Handling Outlier (IQR capping):\n")
5 for col in available_outlier:
6     Q1 = df_clean[col].quantile(0.25)
7     Q3 = df_clean[col].quantile(0.75)
8     IQR = Q3 - Q1
9     lower = Q1 - 1.5 * IQR
10    upper = Q3 + 1.5 * IQR
11    outliers_before = ((df_clean[col] < lower) | (df_clean[col] > upper)).sum()
12    df_clean[col] = df_clean[col].clip(lower=lower, upper=upper)
13    print(f" {col}: {outliers_before:,} outlier di-cap (range: {lower:.2f} - {upper:.2f})")
14
15 print(f"\nOutlier handling completed. Shape: {df_clean.shape}")
```

Handling Outlier (IQR capping):

annual_inc: 19,899 outlier di-cap (range: -20929.50 - 154882.50)
dti: 43 outlier di-cap (range: -5.77 - 39.91)
revol_bal: 22,879 outlier di-cap (range: -14467.00 - 41213.00)
revol_util: 19 outlier di-cap (range: -14.05 - 127.95)
delinq_2yrs: 83,302 outlier di-cap (range: 0.00 - 0.00)

Outlier handling completed. Shape: (466285, 35)

- Feature Encoding on Categorical Variables

```
1 cat_columns = df_clean.select_dtypes(include=['object']).columns.tolist()
2 num_columns = df_clean.select_dtypes(include=['int64', 'float64']).columns.tolist()
3 if 'loan_label' in num_columns:
4     num_columns.remove('loan_label')
5
6 print(f"Categorical Features ({len(cat_columns)}): {cat_columns}")
7 print(f"Numerical Features: {len(num_columns)} Features")
```

```
1 le_dict = {}
2 for col in cat_columns:
3     le = LabelEncoder()
4     df_clean[col] = le.fit_transform(df_clean[col].astype(str))
5     le_dict[col] = le
6     print(f" {col}: {len(le.classes_)} encoded categories")
7
8 print(f"\nShape data after encoding: {df_clean.shape}")
```

term: 2 encoded categories
grade: 7 encoded categories
sub_grade: 35 encoded categories
emp_length: 11 encoded categories
home_ownership: 6 encoded categories
verification_status: 3 encoded categories
issue_d: 91 encoded categories
loan_status: 9 encoded categories
purpose: 14 encoded categories
addr_state: 50 encoded categories
earliest_cr_line: 664 encoded categories
initial_list_status: 2 encoded categories

Shape data after encoding: (466285, 35)

4. DATA PREPARATION

- Split Data & Feature Scaling

The data is divided into a training set (80%) and a testing set (20%), then StandardScaled the numeric features. A stratified split is used to keep the proportion of target labels balanced across both sets.

```
1 from sklearn.model_selection import train_test_split
2
3 # Menyiapkan fitur (X) dan target (y)
4 # Kita drop target 'loan_label' dan target original 'loan_status' jika masih ada
5 cols_to_drop_from_X = ['loan_label', 'loan_status']
6 X = df_clean.drop(columns=[col for col in cols_to_drop_from_X if col in df_clean.columns])
7 y = df_clean['loan_label']
8
9 # Split Data (80:20) dengan Stratify
10 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
11
12 print(f"Dimensi X_train: {X_train.shape}")
13 print(f"Dimensi X_test: {X_test.shape}")
```

Dimensi X_train: (373028, 33)
Dimensi X_test: (93257, 33)

```
1 # Inisialisasi StandardScaler
2 scaler = StandardScaler()
3
4 # Mengambil hanya kolom numerik asli (tidak termasuk kolom hasil One-Hot Encoding yang berisi 0 dan 1)
5 # Jika Anda ingin melakukan scale pada semua fitur X, Anda bisa langsung pakai: num_cols_to_scale = X_train.columns
6 num_cols_to_scale = [col for col in X_train.columns if X_train[col].nunique() > 2]
7
8 # Fit dan Transform pada Data Training
9 X_train[num_cols_to_scale] = scaler.fit_transform(X_train[num_cols_to_scale])
10
11 # HANYA Transform pada Data Testing
12 X_test[num_cols_to_scale] = scaler.transform(X_test[num_cols_to_scale])
13
14 print("Scaling selesai. Data siap dimasukkan ke tahap Data Modelling!")
```

Scaling selesai. Data siap dimasukkan ke tahap Data Modelling!

```
1 df_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 466285 entries, 0 to 466284
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   loan_amnt                             466285 non-null  int64
1   funded_amnt                           466285 non-null  int64
2   funded_amnt_inv                       466285 non-null  float64
3   term                                  466285 non-null  int64
4   int_rate                              466285 non-null  float64
5   installment                           466285 non-null  float64
6   grade                                  466285 non-null  int64
7   sub_grade                             466285 non-null  int64
8   emp_length                            466285 non-null  int64
9   home_ownership                        466285 non-null  int64
10  annual_inc                            466285 non-null  float64
11  verification_status                   466285 non-null  int64
12  issue_d                               466285 non-null  int64
13  loan_status                           466285 non-null  int64
14  purpose                                466285 non-null  int64
15  addr_state                            466285 non-null  int64
16  dti                                    466285 non-null  float64
17  delinq_2yrs                           466285 non-null  float64
18  earliest_cr_line                       466285 non-null  int64
19  inq_last_6mths                         466285 non-null  float64
20  open_acc                               466285 non-null  float64
21  pub_rec                                466285 non-null  float64
22  revol_bal                              466285 non-null  int64
23  revol_util                             466285 non-null  float64
24  total_acc                              466285 non-null  float64
25  initial_list_status                     466285 non-null  int64
26  collections_12_mths_ex_med             466285 non-null  float64
27  acc_now_delinq                          466285 non-null  float64
28  tot_coll_amt                           466285 non-null  float64
29  tot_cur_bal                             466285 non-null  float64
30  total_rev_hi_lim                       466285 non-null  float64
31  loan_label                             466285 non-null  int64
32  has_history_delinq                      466285 non-null  int64
33  has_public_record                       466285 non-null  int64
34  is_currently_delinq                     466285 non-null  int64
dtypes: float64(16), int64(19)
memory usage: 124.5 MB
```

5. DATA MODELING

Logistic Regression

ROC-AUC Score: 0.6826

Classification Report:

	precision	recall	f1-score	support
Bad Loan (0)	0.19	0.63	0.29	11066
Good Loan (1)	0.93	0.63	0.75	82191
accuracy			0.63	93257
macro avg	0.56	0.63	0.52	93257
weighted avg	0.84	0.63	0.7	93257

Decision Tree

ROC-AUC Score: 0.6688

Classification Report:

	precision	recall	f1-score	support
Bad Loan (0)	0.18	0.67	0.28	11066
Good Loan (1)	0.93	0.59	0.72	82191
accuracy			0.6	93257
macro avg	0.55	0.63	0.5	93257
weighted avg	0.84	0.6	0.67	93257

Random Forest

ROC-AUC Score: 0.685

Classification Report:

	precision	recall	f1-score	support
Bad Loan (0)	0.22	0.44	0.29	11066
Good Loan (1)	0.91	0.78	0.84	82191
accuracy			0.74	93257
macro avg	0.56	0.61	0.57	93257
weighted avg	0.83	0.74	0.78	93257

Gradient Boosting

ROC-AUC Score: 0.6978

Classification Report:

	precision	recall	f1-score	support
Bad Loan (0)	0.53	0	0	11066
Good Loan (1)	0.88	1	0.94	82191
accuracy			0.88	93257
macro avg	0.71	0.5	0.47	93257
weighted avg	0.84	0.88	0.83	93257

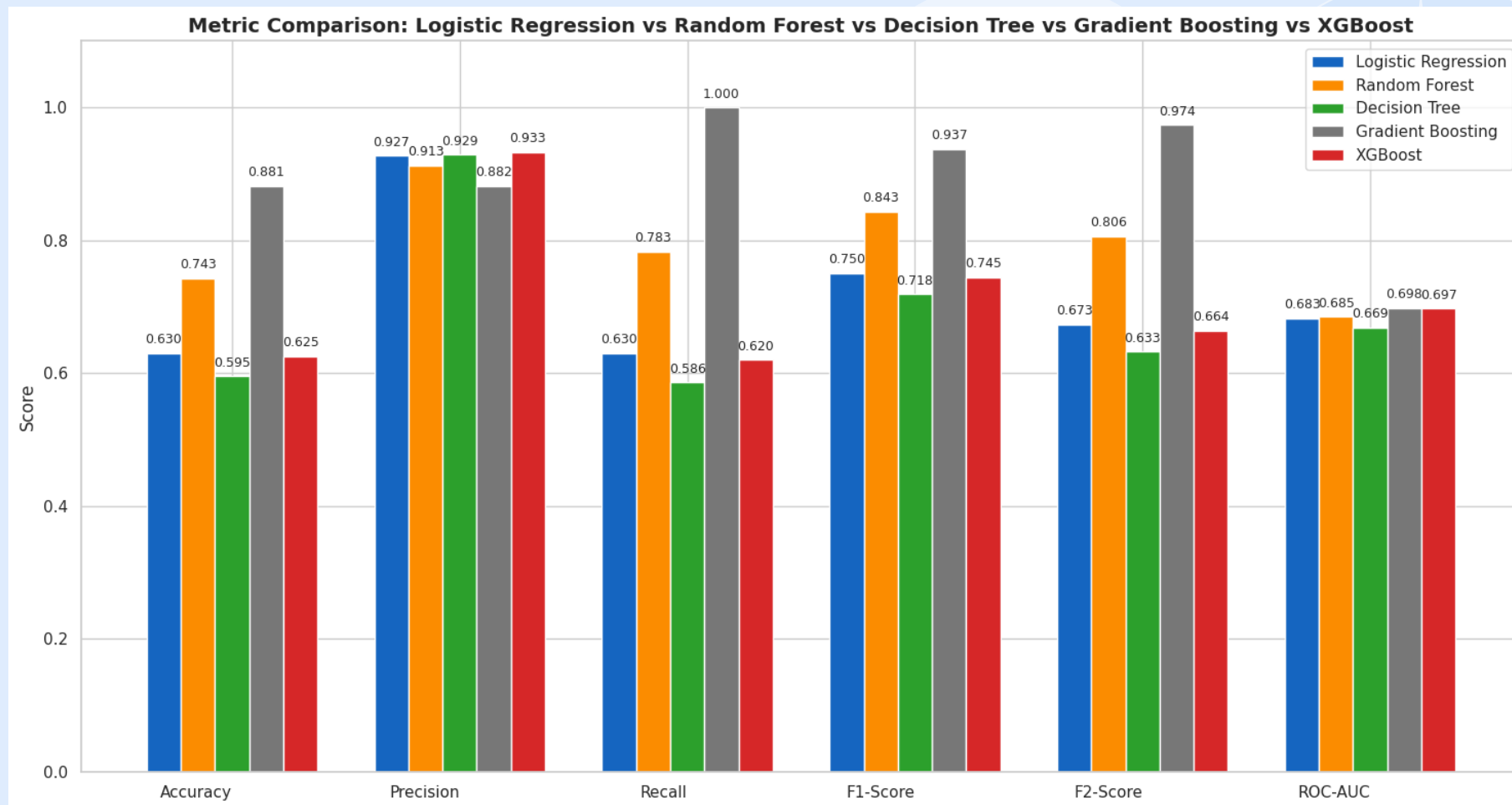
XGBoost

ROC-AUC Score: 0.6972

Classification Report:

	precision	recall	f1-score	support
Bad Loan (0)	0.19	0.67	0.3	11066
Good Loan (1)	0.93	0.62	0.74	82191
accuracy			0.63	93257
macro avg	0.56	0.64	0.52	93257
weighted avg	0.84	0.63	0.69	93257

6. EVALUATION



Business Impact

1. Minimizes financial loss by reducing missed bad loans
2. Supports risk-based pricing and approval decisions
3. Enables explainable and auditable credit decisioning

Final Recommendation

1. Deploy XGBoost with recall-focused thresholding
2. Apply human-in-the-loop review for high-risk loans

Credit Risk Prediction Model

ID/X PARTNERS - DATA SCIENTIST

Presented by
Kuntur Jalassuad